

Finite-moment refinements for SRHT and Khatri–Rao sketches

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Abstract

We refine finite-moment bounds for two structured subspace embeddings by retaining information that is lost in a terminal operator-norm estimate. For two-round SRHT, successive polynomial prefixes use smaller sign cutoffs, yielding an ordered product of nonnegative comparison matrices. For Gaussian Khatri–Rao sketches, an ordered first-return comparison retains the covariance spectrum at the vacuum boundary, and exact fourth-order calibration gives additional finite-sample certificates. We also transfer every even trace-moment bound for Gaussian product probes to independent normalized spherical product probes by conditional Jensen, compute their exact covariance reduction, and give an exact fourth-moment example. A conditional spherical Jacobi comparison explains the related linear-pencil case. The proofs identify the estimates imported from earlier manuscripts and distinguish finite-parameter gains from changes to universal asymptotic constants.

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1 Scope and contributions

For a self-adjoint random matrix Z , an even trace moment gives the sufficient tail certificate

$$\mathbb{P}\{\|Z\| > \varepsilon\} \leq \varepsilon^{-2q} \mathbb{E} \operatorname{Tr} Z^{2q}, \quad q \geq 1. \quad (1.1)$$

Throughout, Tr is the unnormalized trace, $\|\cdot\|$ is the operator norm, and comparison vectors use the Euclidean norm. A sufficient sample count obtained from (1.1) is not asserted to be an optimal threshold.

The common starting point is multiplication by the original random matrix on vector-valued L^2 . The constant functions form the vacuum. A cutoff containing the first q vacuum iterates preserves

the $2q$ -th trace moment. The estimates below keep the probability law of the original SRHT or Gaussian Khatri–Rao sketch; the spherical-product section explicitly introduces a different probe law.

Setting	Information retained	Resulting refinement
Two-round SRHT	Moment-step cutoffs and selector occupation	Ordered-prefix moment and row certificates
Gaussian Khatri–Rao	Covariance eigenvalues and exact fourth-order reuse	Boundary comparison and calibrated fourth moments
Spherical products	Gaussian radius–direction decomposition	All-order transfer and a sharper covariance
Spherical linear pencils	Conditional covariance and Hermite grade costs	Finite Jacobi comparison

Provenance. The original-law multiplication representation, positive-selector estimates, and ordered covariance-spectrum comparison are developed in [1]. The SRHT row-factor estimates come from [2], and the nonlinear Khatri–Rao repeated-visit envelope comes from [3]. The Gaussian creation estimate used here is due to Bandeira, Kunisky, Nizić-Nikolac, Pesenti, and Wang [4]. Gaussian Wick contraction, Gaussian polar decomposition, conditional Jensen, and the Schur-complement renewal identity are classical ingredients. Our purpose is to prove the stated refinements and extensions from these inputs; no exhaustive literature-priority claim is made for the combinations. These new statements have not been formally verified in Lean.

What the numerical comparisons establish. Every displayed row count is tied to a specified moment order, model, and comparison bound. The SRHT table improves a terminal-grade certificate. The covariance-spectrum Khatri–Rao table improves a stationary scalar-boundary certificate, while the fourth-order table compares fourth-order certificates. None of these comparisons alone replaces a universal leading constant in the strongest available all-order theorem.

2 Two-round SRHT: selector returns and ordered prefixes

The refinement in this section changes how the existing local row estimates are assembled. It retains selector grades, then uses the smaller sign spaces reached by successive prefixes of the moment word. The distribution of the sketch, its coherence parameters, and the local coefficient contractions are unchanged. In particular, the argument uses the original sign law.

Let $A, B \in \mathbb{C}^{n \times n}$ be unitary and let $U \in \mathbb{C}^{n \times d}$ satisfy $U^*U = I_d$. Define

$$\alpha = n \max_{r,a} |A_{ra}|^2, \quad \beta = n \max_{a,i} |B_{ai}|^2, \quad W = AD_y B D_x U,$$

where D_x, D_y are independent diagonal matrices of independent unbiased signs. Write $w_r^* = e_r^* W$. Independently choose a uniform m -element subset $T \subseteq [n]$, and set

$$E_T = \frac{n}{m} \sum_{r \in T} w_r w_r^* - I_d. \quad (2.1)$$

For flat unitary transforms, including normalized Walsh transforms, $\alpha = \beta = 1$.

2.1 The finite-moment certificate

Fix an integer $q \geq 1$ and $1 \leq m < n$, and write

$$\rho = m/n, \quad u = \sqrt{(1-\rho)/\rho}, \quad z = |1 - 2\rho|/\rho, \quad \underline{c} = \max \left\{ \frac{1}{2}, 1 - \frac{1}{2\sqrt{m(1-m/n)}} \right\}.$$

For $1 \leq j \leq q$ and $1 \leq k \leq \min(j, n)$, define

$$\Lambda_{j,k} = \left(\sqrt{\alpha k(2j+1)} + \sqrt{2\{\beta(d+8j^2) + \alpha k(2j-1)\}} \right)^2, \quad (2.2)$$

$$\ell_{j,k} = \min\{1, \Lambda_{j,k}/n\}, \quad \ell_{j,0} = 0. \quad (2.3)$$

Let $G^{[j]}$ be the symmetric nonnegative matrix on indices $0, \dots, q$ whose nonzero entries are confined to $0, \dots, \min(j, n)$ and satisfy

$$G_{kk}^{[j]} = z\ell_{j,k}, \quad G_{k,k+1}^{[j]} = G_{k+1,k}^{[j]} = u\sqrt{\ell_{j,k+1}}. \quad (2.4)$$

Here and below e_0 is the first standard basis vector of this scalar comparison space.

Theorem 2.1 (Ordered-prefix SRHT certificate). *Under the preceding assumptions,*

$$\boxed{\mathbb{E} \operatorname{Tr} E_T^{2q} \leq d \underline{c}^{-2q} \|G^{[q]} G^{[q-1]} \dots G^{[1]} e_0\|_2^2} \quad (2.5)$$

Consequently the right-hand side divided by ε^{2q} bounds $\mathbb{P}\{\|E_T\| > \varepsilon\}$. Moreover,

$$\|G^{[q]} \dots G^{[1]} e_0\|_2^2 \leq \|(G^{[q]})^q e_0\|_2^2 = ((G^{[q]})^{2q})_{00}. \quad (2.6)$$

Thus the ordered-prefix certificate is never worse than the terminal-cutoff selector certificate with the same parameters. For $m = n$, $E_T = 0$ exactly. The conclusions also hold when each sign family is $\min(n, 4q)$ -wise independent, provided the two families remain mutually independent and independent of T .

The order in (2.5) matters. The matrices $G^{[j]}$ need not commute, and the squared norm of their product is not generally a vacuum entry of the power of one matrix. The prefactor remains d : the retained sign space does not supply extra initial trace states.

2.2 Local row estimates and the legal cutoff

We record explicitly which local estimates are inherited from the existing proof. In the Boolean character basis of the two sign families, multiplication by x_i is $X_i = a_{x,i} + a_{x,i}^*$, where creation inserts an unoccupied index and annihilation removes an occupied index. In particular, $X_i^* X_i = I$. Sign grades count occupied coordinates.

Let $P_{\text{sig},j}$ retain grades at most $\min(2j, n)$ in each sign family, with the external $\mathbb{C}^d$ factor included. On its range define the positive effects

$$A_r^{[j]} = P_{\text{sig},j} M_{w_r w_r^*} P_{\text{sig},j}, \quad \sum_{r=1}^n A_r^{[j]} = I, \quad (2.7)$$

where M_F denotes pointwise multiplication by the matrix-valued function F . The last identity follows from $W^* W = I_d$ pointwise.

Lemma 2.2 (Subset allowance at a prefix cutoff). *For $S \subseteq [n]$ with $|S| \leq k \leq \min(j, n)$,*

$$\left\| \sum_{r \in S} A_r^{[j]} \right\| \leq \ell_{j,k}. \quad (2.8)$$

Proof. Let $\mathcal{R}_S$ be the row multiplication map $(\mathcal{R}_S f)_r = w_r^* f$, for $r \in S$, with the row label retained as an orthogonal output coordinate. Crucially, its output sign grades are not truncated. Then

$$\sum_{r \in S} A_r^{[j]} = (\mathcal{R}_S P_{\text{sig},j})^* (\mathcal{R}_S P_{\text{sig},j}).$$

Decompose $\mathcal{R}_S = \mathcal{C}_S + \mathcal{A}_S$ according to creation or annihilation in the second sign family, keeping the two first-family transitions joined in $\mathcal{C}_S$.

Here are the precise coefficient estimates used. Set $c_i = e_i^* U$ and $(g_{ri})_a = \sqrt{n} \overline{A_{ra}} B_{ai}$. Unitarity gives

$$\left\| \sum_i t_i g_{ri} \right\|^2 \leq \alpha \sum_i |t_i|^2, \quad \sum_r |g_{ri}\rangle \langle g_{ri}| \preceq \beta I, \quad \sum_i c_i^* c_i = I_d.$$

These inequalities hold with passive Hilbert-space factors. Before a second-family creation, the first inequality and $X_i^* X_i = I$ bound the whole mixed-first-grade coefficient map by $\sqrt{\alpha}$ for each row. Insertion into second-family grade s costs $\sqrt{s+1}$. Orthogonality of the output grades therefore yields

$$n \|\mathcal{C}_S P_{\text{sig},j}\|^2 \leq \alpha k(2j+1). \quad (2.9)$$

For completeness, the fixed-input annihilation estimates inherited from the coefficient-map and Boolean-kernel arguments of [1, 2] are, on input grades (r, s) ,

$$\begin{aligned} n \|\mathcal{R}_S^{+-}\|^2 &\leq \beta(r+1)s, \\ n \|\mathcal{R}_S^{-+}\|^2 &\leq \beta(d+p+pv) + \alpha|S|v, \quad p = (r-1)_+, \quad v = (s-1)_+. \end{aligned}$$

The superscripts are the signs of the changes in the two sign grades. The first estimate factors through a second-family extraction and a first-family insertion. The second is the collective double-annihilation estimate, including the original Boolean occupation correction; this technical kernel estimate is an input from the cited proofs, not a Gaussian replacement or a new assertion here.

For fixed input grades these two outputs are orthogonal. Each fixed output grade under $\mathcal{A}_S$ has at most two possible input grades, so Cauchy–Schwarz contributes a factor of two when mixed input grades are assembled. Since $r, s \leq 2j$ implies $p + pv + (r+1)s \leq 8j^2$, this gives

$$n \|\mathcal{A}_S P_{\text{sig},j}\|^2 \leq 2\{\beta(d+8j^2) + \alpha k(2j-1)\}. \quad (2.10)$$

The triangle inequality applied after (2.9)–(2.10) proves the allowance $\Lambda_{j,k}/n$. Positivity and (2.7) cap it at one. $\square$

This is the existing retained row-factor argument with two parameters kept separate: j controls the sign cutoff, while k controls the number of selected row effects. Substituting k for j throughout (2.2) is unjustified: a low selector grade can coexist with a high sign grade.

2.3 Proof of the moment and sampling comparisons

Proof of Theorem 2.1. First use independent Bernoulli selectors η_r of mean ρ , independent of the signs, and put

$$E_\rho = \rho^{-1} \sum_r \eta_r w_r w_r^* - I_d.$$

In the orthonormal basis $1, (\eta_r - \rho)/\sqrt{\rho(1-\rho)}$, multiplication by $\eta_r/\rho - 1$ is

$$\begin{pmatrix} 0 & u \\ u & \theta \end{pmatrix}, \quad \theta = (1-2\rho)/\rho.$$

Selector grades are therefore subset cardinalities. On the stage- j sign space and selector grades at most $\min(j, n)$, the compressed multiplier has the exact form

$$M_j = u(K_j + K_j^*) + \theta D_j,$$

where

$$(K_j f)_S = \sum_{r \in S} A_r^{[j]} f_{S \setminus \{r\}}, \quad (D_j f)_S = \left(\sum_{r \in S} A_r^{[j]} \right) f_S.$$

We give the positive-effects estimate behind its block bounds. For an output set S of size $k+1$,

$$\left\| \sum_{r \in S} A_r^{[j]} v_r \right\|^2 \leq \left\| \sum_{r \in S} A_r^{[j]} \right\| \sum_{r \in S} \langle v_r, A_r^{[j]} v_r \rangle.$$

This follows by factoring the row through the square roots of the positive effects. Sum over S , noting that each input set J is charged $\sum_{r \notin J} A_r^{[j]} \preceq I$. Lemma 2.2 then gives

$$\|K_j : \text{grade } k \rightarrow k+1\| \leq \sqrt{\ell_{j,k+1}}, \quad \|D_j : \text{grade } k \rightarrow k\| \leq \ell_{j,k}.$$

The reverse edge has the same bound by adjointness. These are precisely the entries of (2.4) after taking absolute values.

Let M denote the uncompressed multiplication operator for E_ρ , and let Ω_a be the constant vector e_a , with both sign families and the selector family in their vacuum states. Multiplication raises each sign grade by at most two and the selector grade by at most one. Hence $f_j = M^j \Omega_a$ lies in the stage- j space, and, with the nested spaces identified isometrically,

$$f_j = M_j f_{j-1}.$$

This identity is asserted for these actual prefixes; it does not assert that compressions preserve arbitrary products. Let $b_j(k)$ be the norm of the selector-grade- k component of f_j , including all its sign coordinates. The block bounds imply componentwise

$$b_j \leq G^{[j]} b_{j-1}, \quad b_0 = e_0.$$

All entries are nonnegative, so iteration and orthogonality give

$$\mathbb{E} \text{Tr } E_\rho^{2q} = \sum_{a=1}^d \|M^q \Omega_a\|^2 \leq d \|G^{[q]} \dots G^{[1]} e_0\|_2^2. \quad (2.11)$$

We now transfer to the original fixed-size law, spelling out the centered coupling because it is distinct from a with-replacement comparison. Draw a Bernoulli subset $\mathcal{B}$ of size $K \sim \text{Bin}(n, \rho)$. If $K > m$, remove uniformly chosen elements until m remain; if $K < m$, add uniformly chosen missing elements. The resulting T is a uniform m -subset and is independent of K . Set

$$a = \mathbb{E}(K - m)_+ = \mathbb{E}(m - K)_+, \quad v = m(1 - m/n).$$

The conditional inclusion probabilities of $\mathcal{B}$ inside and outside T are $1 - a/m$ and $a/(n - m)$, respectively. Consequently, for arbitrary deterministic Hermitian matrices H_r ,

$$Z_{\mathcal{B}} = \sum_r (\mathbf{1}_{r \in \mathcal{B}} - \rho) H_r, \quad Z_T = \sum_r (\mathbf{1}_{r \in T} - \rho) H_r, \quad \mathbb{E}[Z_{\mathcal{B}} \mid T] = c Z_T, \quad c = 1 - a/v.$$

The binomial identity $k \binom{n}{k} = n \binom{n-1}{k-1}$ shows that

$$a = v \mathbb{P}\{\text{Bin}(n-1, \rho) = m\}.$$

The atoms at $m-1$ and m of this latter binomial law are equal, so $c \geq 1/2$. Also $a = \frac{1}{2} \mathbb{E}|K - m| \leq \sqrt{v}/2$, proving $c \geq \underline{c}$. Convexity of $Z \mapsto \text{Tr } Z^{2q} = \|Z\|_{S_{2q}}^{2q}$ on Hermitian matrices and conditional Jensen yield

$$\mathbb{E} \text{Tr } Z_T^{2q} \leq c^{-2q} \mathbb{E} \text{Tr } Z_{\mathcal{B}}^{2q}.$$

Apply this conditionally on the signs with $H_r = w_r w_r^*$, then rescale by ρ^{-1} and use (2.11). This proves (2.5).

For fixed $k \geq 1$, $\Lambda_{j,k}$ is nondecreasing in j . After zero embedding, $0 \leq G^{[j]} \leq G^{[q]}$ entrywise for every $j \leq q$. Nonnegative induction proves (2.6). Finally, $\text{Tr } E_T^{2q}$ has degree at most $4q$ in each sign family. Moment matching therefore transfers the fully independent proof to the stated limited-independent signs. Markov’s inequality and $\|E_T\|^{2q} \leq \text{Tr } E_T^{2q}$ finish the proof. $\square$

2.4 Certified sufficient sample counts

Table 1 keeps the same moment order, sampling transfer, and target failure probability in both columns. It compares the two sides of (2.6), not against a coarser universal scalar prescription.

d	q	Terminal cutoff	Ordered prefixes	Reduction
10	6	28 080	9 928	64.64%
100	8	60 271	26 685	55.72%
1 000	10	169 157	117 011	30.83%
10 000	11	1 011 962	933 686	7.74%

Table 1: Sufficient row counts for $n = 2^{30}$, $\alpha = \beta = 1$, $\varepsilon = 1/2$, $\delta = 1/100$, and $q = \lceil \log_4(4d/\delta) \rceil$. These are certified prescriptions valid for every fixed isometry U , not empirical thresholds or optimal sample counts.

The displayed counts were located by a floating-point search, increased by one percent, and then verified using rational arithmetic. Every square root in a matrix entry was rounded upward to a rational with denominator 10^{18} ; the square root used to lower-bound $\underline{c}$ was rounded downward. The nonnegative vector recurrences were evaluated exactly. For the ordered-prefix column, the certified upper bounds on $d\bar{c}^{-2q}\|G^{[q]}\dots G^{[1]}e_0\|_2^2/(\delta\varepsilon^{2q})$ are, rounded upward, 0.937586, 0.918227, 0.899509, and 0.890415. The exact rational comparisons, rather than these decimal summaries, certify sufficiency; the accompanying `checks/check_srht.py` and `checks/check_srht_terminal.py` reproduce the two columns.

The improvement does not change the local term $\beta(d+8j^2)$, the two-round structure, or the confidence exponent of the underlying theorem. No smaller universal scalar row coefficient is claimed. The new contribution here is the explicit original-law selector certificate and its ordered-prefix refinement; the row contractions, positive-effects mechanism, and centered sampling coupling are inherited from [1, 2].

3 Khatri–Rao: covariance-sensitive boundaries and fourth-order calibration

Fix an isometry $U : \mathbb{R}^r \rightarrow \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}$. For independent standard Gaussian vectors g_i, h_i , put

$$y_i = U^T(g_i \otimes h_i), \quad X_i = y_i y_i^T - I_r, \quad Z = \frac{1}{m} \sum_{i=1}^m X_i, \quad W = \mathbb{E} X_1^2.$$

Thus $\mathbb{E} X_i = 0$, and $\|Z\| \leq \varepsilon$ is the usual squared-norm embedding guarantee on the fixed subspace $\text{ran } U$. All traces below are unnormalized. The coefficients and the subspace are fixed before the sketch is drawn.

The original Khatri–Rao proof already propagates ordered, stage-dependent comparison matrices and permits the exact value of $\|W\|$ in its activation bounds [3]. Our additional stationary comparison retains the entire spectrum of W at the vacuum boundary. A separate fourth-order calculation retains the exact contribution of one original sample and improves the Gaussian pairing contribution. Neither argument identifies the quartic Gaussian polynomial X_i with a linear Gaussian matrix series.

3.1 An ordered first-return comparison

We record the trace comparison used below, including its proof to make the noncommutative step explicit. This is the covariance-spectrum comparison of [1, Theorem 4.3 and Corollary 4.4]; the underlying renewal identity is the classical block-resolvent expansion.

Lemma 3.1 (Covariance at the boundary). *Let M be a finite-dimensional self-adjoint operator with decomposition*

$$M = \begin{pmatrix} 0 & B^* \\ B & C \end{pmatrix}, \quad V = B^*B,$$

relative to a vacuum space $\mathcal{H}_0$ and its orthogonal complement. Decompose the complement into sectors $\mathcal{H}_1, \dots, \mathcal{H}_K$, assume $\text{ran } B \subseteq \mathcal{H}_1$, and suppose the symmetric entrywise nonnegative matrix G satisfies $\|P_i C P_j\| \leq G_{ij}$. For $v \geq 0$, set

$$T(v) = \begin{pmatrix} 0 & \sqrt{v} e_1^\top \\ \sqrt{v} e_1 & G \end{pmatrix}.$$

For every integer $q \geq 1$,

$$\text{Tr}_{\mathcal{H}_0}(P_0 M^{2q} P_0) \leq \sum_{\lambda \in \text{spec } V} (T(\lambda)^{2q})_{00}. \quad (3.1)$$

Eigenvalues are counted with multiplicity. If $s \geq \|V\|$ and $s > 0$, the right side is at most $(\text{Tr } V/s)(T(s)^{2q})_{00}$.

Proof. Put $K_j = P_0 M^j P_0|_{\mathcal{H}_0}$ and $F_j = B^* C^{j-2} B$ for $j \geq 2$. Resolvent expansion, or splitting a vacuum walk at its first return, gives

$$K_0 = I, \quad K_1 = 0, \quad K_n = \sum_{j=2}^n F_j K_{n-j}.$$

The order of the factors is retained. Write the polar decomposition $B = QV^{1/2}$. Then $F_j = V^{1/2} A_j V^{1/2}$, where $A_j = Q^* C^{j-2} Q$ and $\|A_j\| \leq c_j := (G^{j-2})_{11}$. The last inequality follows by block-norm propagation inside the complement; Q is a contraction with range in sector one. For any composition $j_1 + \dots + j_\ell = n$ with $j_t \geq 2$, cyclicity and Schatten Hölder give

$$|\text{Tr}(F_{j_1} \dots F_{j_\ell})| = |\text{Tr}(A_{j_1} V \dots A_{j_\ell} V)| \leq \left(\prod_{t=1}^{\ell} c_{j_t} \right) \text{Tr } V^\ell.$$

For the scalar matrix $T(v)$ the corresponding first-return weight is vc_j . Expanding both renewals and summing the preceding estimate proves (3.1). In particular, $f_q(v) = (T(v)^{2q})_{00}$ is a polynomial with nonnegative coefficients and zero constant term. For $0 \leq v \leq s$, $f_q(v) \leq (v/s)f_q(s)$, proving the last assertion. No Loewner substitution inside ordered products is used. $\square$

3.2 The original-law Khatri–Rao certificate

For $q \geq 1$, set $K = \min(q, m)$ and

$$\mathcal{B}_q(r) = e(27rq + 192q^2 + 24q).$$

Choose any certified $\nu \geq \|W\|$; the original Wick estimate permits $\nu = 4r + 4$. For $v \geq 0$, define a symmetric tridiagonal matrix $T_{q,m}^\nu(v)$ on $0, \dots, K$ by

$$\begin{aligned} (T_{q,m}^\nu(v))_{00} &= 0, & (T_{q,m}^\nu(v))_{kk} &= \mathcal{B}_q(r)/m \quad (1 \leq k \leq K), \\ (T_{q,m}^\nu(v))_{01} &= (T_{q,m}^\nu(v))_{10} = \sqrt{v}, & (T_{q,m}^\nu(v))_{k,k+1} &= \frac{\sqrt{(k+1)(m-k)\nu}}{m} \quad (1 \leq k < K). \end{aligned} \quad (3.2)$$

The symmetric counterparts of the last entries are understood. The positive diagonal allowance accounts for repeated visits to an already active sample.

Theorem 3.2 (Covariance-spectrum Khatri–Rao bound). *Let $\lambda_1, \dots, \lambda_r$ be the eigenvalues of W . Then*

$$\mathbb{E} \operatorname{Tr} Z^{2q} \leq \sum_{a=1}^r (T_{q,m}^\nu(\lambda_a/m)^{2q})_{00} \quad (3.3)$$

$$\leq \frac{\operatorname{Tr} W}{\nu} (T_{q,m}^\nu(\nu/m)^{2q})_{00}. \quad (3.4)$$

Consequently, either right side being at most $\delta \varepsilon^{2q}$ is sufficient for $\mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta$.

Proof. Work in $\mathbb{R}^r \otimes \bigotimes_{i=1}^m L^2(g_i, h_i)$ and let $\mathcal{M}_Z$ be multiplication by Z on vector polynomials. Restrict to functions even under each separate sign flip of g_i and h_i . Compress to total Hermite degree at most $4q$ and at most q nonconstant sample registers. Denote the projection by Π_q and the compressed operator by $M_q = \Pi_q \mathcal{M}_Z \Pi_q$. This is a finite-dimensional symmetric matrix. Multiplication raises total degree by at most four and activates at most one sample. Thus, for every vacuum vector v and $j \leq q$, $M_q^j v = \mathcal{M}_Z^j v$, and therefore

$$\mathbb{E} \operatorname{Tr} Z^{2q} = \operatorname{Tr}_{\mathbb{R}^r} (P_0 M_q^{2q} P_0).$$

This uses only finite polynomial moments; the full Gaussian multiplier need not be bounded.

Grade by the number of nonconstant registers. Centering gives a zero vacuum block. Distinct samples applied to vacuum are orthogonal, so its first-step Gram is exactly $B^*B = W/m$, and B lands in grade one. The fresh-sample contraction of [3, Chain C, “Fresh-sample contraction”] gives

$$\|P_{k+1} M_q P_k\| \leq \frac{\sqrt{(k+1)(m-k)\nu}}{m}.$$

For completeness, each missing sample contributes a map with squared norm at most ν/m^2 ; an output set of size $k+1$ receives $k+1$ inputs, and each input set has $m-k$ missing labels. Cauchy–Schwarz and orthogonality give the displayed estimate on the full grade, not just on a basis vector.

For the diagonal block we import exactly the proved original-law bound [3, Chain C, “Sum of already-active sample interactions”]: $\|P_k M_q P_k\| \leq \mathcal{B}_q(r)/m$ for $k \geq 1$. Its inputs are the local-degree estimate

$$\|y^\top f_d\|_2^2 \leq e[(2d+1)r + 4d^2 + 2d] \|f_d\|_2^2$$

and the fact that at most three input degrees overlap at any output degree. Summing over the actual local degrees d_i , using $\sum d_i \leq 4q$, $\sum d_i^2 \leq 16q^2$, and at most q active registers, gives $\mathcal{B}_q(r)$. Centering contributes a lower bound $-qI$, also covered by $\mathcal{B}_q(r)$. This retains the nonlinear repeated-sample costs of the original proof. Apply Lemma 3.1 to the rest matrix specified in (3.2), then use trace Markov. $\square$

If $W = sI_r$ and $\nu = s$, (3.3) is exactly the old scalar stationary certificate with exact variance. Thus flat covariance does not illustrate an additional spectrum gain. A nonflat example does. Represent $y_a = g^\top B_a h$ using

$$B_1 = \operatorname{diag}(a, b), \quad B_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad a^2 + b^2 = 1, \quad ab = \frac{1}{12}.$$

These matrices are Frobenius-orthonormal. Wick’s formula gives $\mathbb{E} y_1^4 = 107/12$, $\mathbb{E} y_2^4 = 6$ and $\mathbb{E} y_1^2 y_2^2 = 19/6$; the off-diagonal covariance vanishes by sign symmetry. Hence

$$W = \operatorname{diag}(133/12, 49/6), \quad \|W\| = 133/12, \quad \frac{\operatorname{Tr} W}{\|W\|} = \frac{231}{133} < 2.$$

At $q = 3$, $\varepsilon = 1/2$, and $\delta = 1/100$, the following are sufficient row counts for the same stationary repeated-visit allowance:

Activation/boundary information retained	Sufficient m
Uniform allowance $4r + 4$ at all edges	10 581
Exact norm $\ W\ $ at all edges	10 403
Exact spectrum at the boundary, exact norm inside	10 107

The accompanying rational verification checks each displayed integer and failure of its predecessor. The certificate decreases for $m \geq 2q$: $1/m$ decreases, and each squared interior edge is proportional to $(m - k)/m^2$, whose derivative is $(2k - m)/m^3 \leq 0$ in this range. These are comparisons of stationary certificates: the original manuscript also has stronger ordered degree-allocation certificates, whose minimum with (3.3) remains a valid choice.

3.3 Exact fourth-order calibration

This next statement applies beyond Khatri–Rao. Let X be a centered real symmetric random matrix with finite fourth moment, let $X_1, \dots, X_m$ be independent copies, and put $Z = m^{-1} \sum_i X_i$. Choose a centered Gaussian symmetric matrix $G = \sum_{\ell=1}^N \gamma_\ell A_\ell$ whose complete matrix-entry covariance matches that of X . Such a representation exists by factoring the covariance on the finite-dimensional real symmetric matrix space. Then

$$W = \mathbb{E} X^2 = \sum_{\ell} A_\ell^2, \quad \kappa_G^2 = \lambda_{\max}([A_j A_i]_{i,j=1}^N).$$

The block matrix here is symmetric, even though its individual blocks need not be. It is the crossed coefficient kernel appearing in the Gaussian creation estimates of Bandeira et al. [4].

Proposition 3.3 (Original-component fourth-order certificate). *With $\mu_4 = \mathbb{E} \operatorname{Tr} X^4$,*

$$\mathbb{E} \operatorname{Tr} Z^4 = \frac{\mu_4 + (m-1) \mathbb{E} \operatorname{Tr} G^4}{m^3} \tag{3.5}$$

$$\leq \frac{\mu_4 + (m-1)(2 \operatorname{Tr} W^2 + \kappa_G^2 \operatorname{Tr} W)}{m^3}. \tag{3.6}$$

Proof. Centering and independence eliminate words with a singleton sample label. There are m all-equal contributions. The words with two distinct labels give $m(m-1)$ times the three covariance pairings, precisely the Gaussian fourth moment. This proves (3.5), the trace specialization of the fourth-order calibration in [1, Lemma 8.2]. Wick expansion gives

$$\mathbb{E} \operatorname{Tr} G^4 = 2 \operatorname{Tr} W^2 + \sum_{i,j} \operatorname{Tr}(A_i A_j A_i A_j).$$

Let S be the block column with entries A_i and let $C = [A_j A_i]_{i,j}$. The last sum is $\operatorname{Tr}(S^* C S)$, which is at most $\kappa_G^2 \operatorname{Tr}(S^* S) = \kappa_G^2 \operatorname{Tr} W$. This proves the bound directly. $\square$

The baseline Gaussian bound is $3 \operatorname{Tr} W^2$; consequently (3.6) improves it when $\kappa_G^2 \operatorname{Tr} W < \operatorname{Tr} W^2$. The original component term μ_4 remains exact. Matching the covariance of the row y instead of the centered matrix $yy^\top - I$ would not justify this proposition.

3.4 An analytic full-product benchmark

Take the whole product space, $y = g \otimes h$ with $g \sim N(0, I_a)$ and $h \sim N(0, I_b)$. Put $r = ab$ and $L = (a+2)(b+2)$. Then

$$W = (L-1)I_r, \quad \mathbb{E} \operatorname{Tr} G^4 = r(2L^2 + 3L + 3). \tag{3.7}$$

To verify the second formula, let X' be an independent copy of X . Covariance matching and Wick expansion imply

$$\mathbb{E} \operatorname{Tr} G^4 = 3 \operatorname{Tr} W^2 - \frac{1}{2} \mathbb{E} \| [X, X'] \|_F^2.$$

For rank-one matrices $P = yy^\top$ and $P' = y'y'^\top$,

$$\frac{1}{2} \| [P, P'] \|_F^2 = \|y\|^2 \|y'\|^2 \langle y, y' \rangle^2 - \langle y, y' \rangle^4.$$

For independent d -dimensional standard Gaussian vectors u, u' , direct conditioning yields

$$\mathbb{E} \|u\|^2 \|u'\|^2 \langle u, u' \rangle^2 = d(d+2)^2, \quad \mathbb{E} \langle u, u' \rangle^4 = 3d(d+2).$$

The two factors separate, so the expected half-commutator square is $rL(L-9)$. Substitution proves (3.7).

We also give a small analytic certificate for the crossed kernel, so the intermediate comparison does not depend on a floating-point eigenvalue. Let $\{A_i^{(d)}\}$ be the GOE coefficient family $\sqrt{2}E_{jj}$ and $E_{jk} + E_{kj}$ for $j < k$. A Gaussian covariance match for X consists of the three coefficient families

$$A_i^{(a)} \otimes I_b, \quad I_a \otimes A_j^{(b)}, \quad A_i^{(a)} \otimes A_j^{(b)}.$$

Indeed, writing $gg^\top = I_a + A$ and $hh^\top = I_b + B$ gives $X = A \otimes I_b + I_a \otimes B + A \otimes B$; the three summands are pairwise uncorrelated, and A, B have GOE covariance. Thus block-norm comparison gives

$$\kappa_G^2 \leq \lambda_{\max} \begin{pmatrix} 2 & \sqrt{(a+1)(b+1)} & 2\sqrt{b+1} \\ \sqrt{(a+1)(b+1)} & 2 & 2\sqrt{a+1} \\ 2\sqrt{b+1} & 2\sqrt{a+1} & 4 \end{pmatrix}. \quad (3.8)$$

Here are the needed block norms. The GOE row map has squared norm $d+1$ because $\sum_i (A_i^{(d)})^2 = (d+1)I_d$. Its crossed kernel, identified with an operator on tensors $u_{ij,k} = u_{ji,k}$, sends $u_{ij,k}$ to $u_{ik,j} + u_{jk,i}$, and hence has norm at most two by the triangle inequality for coordinate permutations. The three diagonal blocks therefore have norms at most 2, 2, 4. The first off-diagonal block is a product of two row/column maps, while each remaining block is the tensor product of one crossed kernel and one row/column map. Their norms are the off-diagonal allowances in (3.8). Applying the scalar Rayleigh bound to the norms of the three vector blocks proves the claim.

For $a = b = 2$, $r = 4$, $L = 16$, and

$$W = 15I_4, \quad \kappa_G^2 \leq \frac{9 + \sqrt{97}}{2}, \quad \mathbb{E} \operatorname{Tr} G^4 = 2252.$$

The kernel bound follows by separating the antisymmetric vector $(1, -1, 0)$ from the remaining two-dimensional block of (3.8). Writing $t = \|y\|^2$, the rank-one identity gives

$$\operatorname{Tr}(yy^\top - I_4)^4 = t^4 - 4t^3 + 6t^2 - 4t + 4.$$

Since $\mathbb{E} t^j = \prod_{k=0}^{j-1} (2+2k)^2$, this yields $\mu_4 = 138612$. Therefore the exact fourth trace moment is

$$\mathbb{E} \operatorname{Tr} Z^4 = \frac{138612 + 2252(m-1)}{m^3}. \quad (3.9)$$

At $\varepsilon = 1/2$, $\delta = 1/100$, fourth-order trace Markov gives:

Gaussian pairing allowance	Value	Sufficient m
Generic $3 \operatorname{Tr} W^2$	2700	2104
Crossed-kernel certificate	$1800 + 30(9 + \sqrt{97})$	1974
Exact covariance pairing	2252	1928

The attached verifier uses rational upper and lower bounds on $\sqrt{97}$ and checks both the stated count and its predecessor. These improvements are respectively 6.18% and 8.37% relative to the generic fourth-order certificate. They do not assert improvement over an optimized all-order certificate, or a new universal constant in the Khatri–Rao embedding theorem. The exact pairing formula is a direct covariance calculation; the comparison with the crossed kernel records explicitly what the combined operator input contributes.

4 Independent spherical-product rows

The Gaussian Khatri–Rao guarantees extend to a different, bounded row distribution by conditional expectation. This extension uses independent rows; the rows of a single Haar frame are not independent instances of this model.

Fix positive integers a, b, r and an isometry $U \in \mathbb{R}^{ab \times r}$, $U^\top U = I_r$. For each sample $i = 1, \dots, m$, let $u_i \in S^{a-1}$ and $v_i \in S^{b-1}$ be independent uniform unit vectors, independent also across samples. Define

$$y_i^{\text{sph}} = U^\top(\sqrt{a} u_i \otimes \sqrt{b} v_i), \quad Z_{\text{sph}} = \frac{1}{m} \sum_{i=1}^m (y_i^{\text{sph}} (y_i^{\text{sph}})^\top - I_r).$$

Let $y_i^{\text{gau}} = U^\top(g_i \otimes h_i)$ and Z_{gau} denote the corresponding rows and centered empirical covariance for independent $g_i \sim N(0, I_a)$ and $h_i \sim N(0, I_b)$. Both row models are isotropic on $\mathbb{R}^r$.

Theorem 4.1 (Convex transfer and exact spherical covariance). *For every convex function F on real symmetric $r \times r$ matrices for which the expectations exist,*

$$\mathbb{E}F(Z_{\text{sph}}) \leq \mathbb{E}F(Z_{\text{gau}}). \quad (4.1)$$

In particular, for every integer $q \geq 1$,

$$\mathbb{E} \text{Tr} Z_{\text{sph}}^{2q} \leq \mathbb{E} \text{Tr} Z_{\text{gau}}^{2q}. \quad (4.2)$$

For a single row, write $W_\star = \mathbb{E}(y^\star (y^\star)^\top - I_r)^2$. Then, with $c_{a,b} = ab/((a+2)(b+2))$,

$$W_{\text{sph}} = c_{a,b}(W_{\text{gau}} + I_r) - I_r. \quad (4.3)$$

Proof. Gaussian polar decomposition gives a coupling

$$g_i = R_i u_i, \quad h_i = S_i v_i, \quad y_i^{\text{gau}} = \sqrt{\tau_i} y_i^{\text{sph}}, \quad \tau_i = \frac{R_i^2 S_i^2}{ab},$$

where $R_i^2 \sim \chi_a^2$, $S_i^2 \sim \chi_b^2$, and all radii are independent of all directions. Since $\mathbb{E}\tau_i = 1$,

$$\mathbb{E}[Z_{\text{gau}} \mid (u_i, v_i)_{i=1}^m] = Z_{\text{sph}}.$$

Conditional Jensen proves (4.1). For symmetric A , $\text{Tr} A^{2q} = \|A\|_{S_{2q}}^{2q}$ is convex, which proves (4.2).

Finally, $\mathbb{E}\tau_i^2 = (a+2)(b+2)/(ab)$ and isotropy give

$$W_{\text{gau}} + I_r = \mathbb{E}(y^{\text{gau}} (y^{\text{gau}})^\top)^2 = c_{a,b}^{-1} \mathbb{E}(y^{\text{sph}} (y^{\text{sph}})^\top)^2 = c_{a,b}^{-1} (W_{\text{sph}} + I_r).$$

□

Corollary 4.2 (Transfer of moment-based embedding certificates). *If a Gaussian Khatri–Rao certificate proves $\mathbb{E} \text{Tr} Z_{\text{gau}}^{2q} \leq C_q(m)$ and $C_q(m) \leq \delta \varepsilon^{2q}$, then the spherical-product sketch satisfies*

$$\mathbb{P}(\|Z_{\text{sph}}\| > \varepsilon) \leq \delta.$$

Consequently, every sufficient row count obtained by this moment method, including the refinements above, remains valid for independent spherical-product rows.

Proof. For a symmetric matrix A , $\|A\|^{2q} \leq \text{Tr} A^{2q}$. Apply Markov's inequality and Theorem 4.1. □

The transfer concerns moment certificates, not a comparison of the two failure probabilities at every threshold. It introduces a new row model without worsening those certificates. Although (4.3) also gives $W_{\text{sph}} \preceq W_{\text{gau}}$, substituting this smaller covariance into every Gaussian interior block estimate would require a new proof: the multiplication laws have changed. The argument uses classical radius–direction independence and Jensen's inequality; the Gaussian certificates being transferred are those of [3] and the present refinements.

4.1 An exact fourth-moment certificate for the full product space

For the full product space, fixed spherical radii permit a stronger calculation than simply transferring the Gaussian certificate.

Proposition 4.3 (Full-product spherical fourth moment). *Let $r = ab$, $L = (a+2)(b+2)$, $U = I_r$, and $x = yy^\top - I_r$ for one spherical-product row. Then*

$$W_{\text{sph}} = (r-1)I_r, \quad \beta_{a,b} := \mathbb{E} \text{Tr} x^4 = (r-1)^4 + r - 1.$$

Define

$$\gamma_{a,b} = 3r(r-1)^2 - r^3 \left(1 - \frac{9}{L}\right).$$

For every positive integer m ,

$$\mathbb{E} \text{Tr} Z_{\text{sph}}^4 = \frac{\beta_{a,b} + (m-1)\gamma_{a,b}}{m^3}. \quad (4.4)$$

Proof. Since $\|y\|^2 = r$, the eigenvalues of x are $r-1$ once and -1 with multiplicity $r-1$. This proves the value of $\beta_{a,b}$; isotropy and $(yy^\top)^2 = ryy^\top$ prove the covariance. Let x' and y' be independent copies. Put

$$\Delta = \frac{1}{2} \mathbb{E} \| [x, x'] \|_F^2 = \text{Tr} W_{\text{sph}}^2 - \mathbb{E} \text{Tr}(xx'xx').$$

The rank-one commutator identity gives

$$\Delta = r^2 \mathbb{E} \langle y, y' \rangle^2 - \mathbb{E} \langle y, y' \rangle^4.$$

For independent uniform unit vectors $u, u' \in S^{a-1}$,

$$\mathbb{E} \langle \sqrt{a}u, \sqrt{a}u' \rangle^2 = a, \quad \mathbb{E} \langle \sqrt{a}u, \sqrt{a}u' \rangle^4 = \frac{3a^3}{a+2}.$$

These standard spherical moments follow by conditioning on u' and using rotational invariance, or by Gaussian polar decomposition. Multiplication of the two independent factor moments yields $\mathbb{E} \langle y, y' \rangle^2 = r$ and $\mathbb{E} \langle y, y' \rangle^4 = 9r^3/L$. Thus $\Delta = r^3(1 - 9/L)$.

In expanding the fourth trace moment of a sum of independent centered x_i , only one-name words and two-name pairings survive. Cyclicity of trace gives

$$\mathbb{E} \text{Tr} \left(\sum_{i=1}^m x_i \right)^4 = m\beta_{a,b} + m(m-1) (2 \text{Tr} W_{\text{sph}}^2 + \mathbb{E} \text{Tr}(xx'xx')).$$

The bracket is $3 \text{Tr} W_{\text{sph}}^2 - \Delta = \gamma_{a,b}$. Division by m^4 proves the result. $\square$

For $a = b = 2$, these quantities are $r = 4$, $W_{\text{sph}} = 3I_4$, $\beta_{2,2} = 84$, and $\gamma_{2,2} = 80$. Therefore

$$\mathbb{E} \text{Tr} Z_{\text{sph}}^4 = \frac{80m+4}{m^3}. \quad (4.5)$$

At $\varepsilon = 1/2$ and $\delta = 1/100$, the smallest integer satisfying the resulting fourth-order Markov certificate is $m = 358$:

$$1600(80 \cdot 358 + 4) \leq 358^3, \quad 1600(80 \cdot 357 + 4) > 357^3.$$

The certificate decreases with m . The corresponding Gaussian fourth-moment certificate in the preceding section requires $m = 1928$. This comparison quantifies the benefit of the spherical-product model for this particular sufficient certificate; it is not a reduction of the Gaussian theorem's constant or a comparison with the best available all-order certificate.

5 Reproducibility and use of the bounds

The accompanying `checks` directory uses only the Python standard library. The script `run_checks.py` runs the SRHT ordered-prefix and terminal comparisons, the Khatri–Rao covariance-spectrum recurrence, the Gaussian fourth-order certificates, and the spherical fourth-order calculation. Floating-point searches locate SRHT candidates; every reported sufficiency claim is then checked by rational upper bounds. The Khatri–Rao verifiers additionally check the preceding integer for the specified monotone certificate. The spherical example is checked both by exact integer threshold inequalities and by an independent explicit covariance basis.

These numerical checks verify evaluation of the displayed certificates. The probabilistic and operator comparisons are justified by the proofs and the explicitly stated imported lemmas. For any fixed application, one may take the minimum of independently valid sufficient moment bounds. This applies, in particular, to the new stationary Khatri–Rao bound and the earlier degree-allocated bounds, and to the transferred Gaussian certificate and the direct spherical fourth-order certificate.

A Conditional spherical linear pencils

For a linear matrix pencil, Gaussian polar decomposition gives an exact moment correction rather than only the convex comparison of Section 4. This is a complementary extension: the centered Khatri–Rao covariance error is nonlinear in its factors and is not itself the linear pencil considered here. Fresh spherical directions in conditional frame arguments, such as those in [5], provide one setting where the following hypotheses apply.

Theorem A.1 (Conditional spherical Jacobi comparison). *Condition on a sigma-field $\mathcal{F}$. Let u be conditionally uniform on the real unit sphere S^{h-1} and let $B_1, \dots, B_h$ be $\mathcal{F}$ -measurable Hermitian $D \times D$ matrices. Set*

$$Y = \sqrt{h} \sum_{i=1}^h u_i B_i, \quad W = \sum_i B_i^2 \neq 0, \quad s = \|W\|, \quad \kappa^2 = \lambda_{\max}([B_j B_i]_{i,j=1}^h).$$

For $p \geq 1$, let $J_p(v)$ be the symmetric tridiagonal matrix on $0, \dots, p$ with zero diagonal and edges

$$[J_p(v)]_{0,1} = \sqrt{v}, \quad [J_p(v)]_{k,k+1} = \sqrt{s + k\kappa^2} \quad (1 \leq k < p).$$

With $a_{h,p} = \prod_{j=0}^{p-1} h/(h+2j)$, almost surely,

$$\begin{aligned} \mathbb{E}[\mathrm{Tr} Y^{2p} \mid \mathcal{F}] &\leq a_{h,p} \sum_{\lambda \in \mathrm{spec}(W)} (J_p(\lambda)^{2p})_{00} \\ &\leq a_{h,p} \frac{\mathrm{Tr} W}{s} (J_p(s)^{2p})_{00}. \end{aligned} \tag{A.1}$$

Eigenvalues are counted with multiplicity. At fourth order the sharper bound

$$\mathbb{E}[\mathrm{Tr} Y^4 \mid \mathcal{F}] \leq \frac{h}{h+2} (2 \mathrm{Tr} W^2 + \kappa^2 \mathrm{Tr} W) \tag{A.2}$$

also holds. If $W = 0$, then $Y = 0$ and the conclusions are interpreted as zero without the displayed division by s .

Proof. Fix the conditioned coefficients and take independent standard real Gaussians g_i . Set $X = \sum_i g_i B_i$. The decomposition $g = Ru$, with R independent of u and $R^2 \sim \chi_h^2$, gives the matrix identity

$$\mathbb{E}[Y^{2p} \mid \mathcal{F}] = a_{h,p} \mathbb{E}_g X^{2p}. \tag{A.3}$$

For completeness, the interior Gaussian bound used in (A.1) is the bosonic specialization of the creation estimate in [4, Proposition 4.1]. On Gaussian Hermite space write $L = \sum_i B_i \otimes a_i^*$, where a_i^* and a_i are creation and annihilation. If $R_k f = (a_i f)_i$ on degree k , the canonical commutation relation gives

$$L^* L = W \otimes I + R_k^* [B_j B_i]_{i,j} R_k, \quad R_k^* R_k = kI.$$

Consequently $\|L\|_{\text{degree } k}^2 \leq s + k\kappa^2$. The degree- p cutoff of $L + L^*$ preserves its first p vacuum iterates, so its vacuum moment of order $2p$ equals $\mathbb{E}_g X^{2p}$. The vacuum-to-degree-one Gram is exactly W . The covariance-spectrum comparison of Lemma 3.1, equivalently [1, Corollary 4.4], therefore bounds this Gaussian trace moment by the sum in (A.1). Its nonnegative return-polynomial relaxation gives the second inequality. Multiplication by $a_{h,p}$ proves both spherical comparisons.

For the fourth-order refinement, Wick’s formula gives

$$\mathbb{E}_g \text{Tr } X^4 = 2 \text{Tr } W^2 + \sum_{i,j} \text{Tr}(B_i B_j B_i B_j).$$

Regarding $(B_i)_i$ as a vector in the direct sum of Hilbert–Schmidt spaces, its quadratic form against $[B_j B_i]_{i,j}$ is the last sum. It is at most $\kappa^2 \sum_i \text{Tr } B_i^2 = \kappa^2 \text{Tr } W$. Apply (A.3) at $p = 2$. $\square$

The fourth-order right side improves the $p = 2$ Jacobi right side by $\frac{h}{h+2}(s \text{Tr } W - \text{Tr } W^2) \geq 0$. The spherical factor is exact and equals one at second order. Its role is to extend a Gaussian comparison to the stated conditional law; it does not improve a proof already using exact conditional second moments. The ingredients have separate origins: classical polar decomposition supplies the radial factor, [4] supplies the interior creation inequality, and [1] supplies the covariance-sensitive boundary comparison. No joint-frame or accumulated adaptive-process bound follows merely from the one-direction conditional statement.

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